

Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE 227: Data Communication

Batch: 24th

Date: 20/03/2025

Final Examination

Semester: Fall 2024

Time: 2 Hours

Total Marks: 40

Answer any **five** questions:

$5 \times 8 = 40$

CLOs Marks

CLO 5 4

CLO 5 4

CLO 4 4

CLO 2 4

CLO 5 4

CLO 5 4

CLO 5 4

CLO 3 4

CLO 1 4

CLO 3 4

CLO 2 4

CLO 2 4

CLO 2 4

7. (a) Draw the constellation diagram for the following:

I. ASK, with peak amplitude values of 1 and 3

II. BPSK, with a peak amplitude value of 2

III. QPSK, with a peak amplitude value of 3

IV. 8-QAM with two different peak amplitude values, 1 and 3, and four different phases.

(b) We need to use synchronous TDM and combine 20 digital sources, each of 100 Kbps. Each output slot carries 1 bit from each digital source, but one extra bit is added to each frame for synchronization. Answer the following questions:

I. What is the size of an output frame in bits?

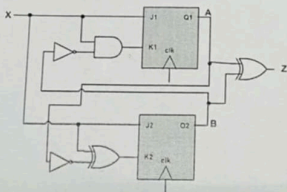
II. What is the output frame rate?

III. What is the duration of an output frame?

IV. What is the output data rate?

20, 10, 20, 100, 12
 200, 600

5. (a) Explain the difference between Moore and Mealy state machines. CLO3 3
 (b) Design a 2-bit sequential circuit that can run in forward ($11 \rightarrow 00 \rightarrow 10 \rightarrow 01 \rightarrow 11 \rightarrow \dots$) or reverse ($11 \rightarrow 01 \rightarrow 10 \rightarrow 00 \rightarrow 11 \rightarrow \dots$) depending on a direction control input, using SR flip-flops. CLO5 5
6. (a) Explain the design procedure for clocked sequential circuits. CLO1 2
 (b) Explain the importance of synchronization in clocked sequential circuits. CLO1 2
 (c) From the given sequential circuit, analyze the possible state transitions and output sequences: CLO4 4



7. (a) How does a PLA differ from a PAL in terms of its architecture and functionality? CLO4 2
 (b) Given the Boolean functions $Y_1 = AB + C$ and $Y_2 = A'C + BC$, design a PAL that implements these functions. CLO5 3
 (c) Design the block diagram of an 8x8 ROM and explain its working. CLO5 3



HAMMARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE225: Digital Logic Design

Batch: 24th

Date: 24/03/2025

Final Examination

Semester: Fall 2024

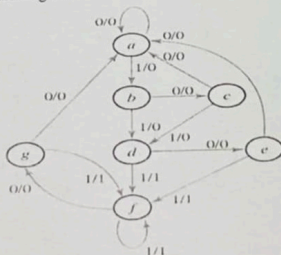
Time: 2 Hours

Total Marks: 40

Answer any **five** questions of the following.

$5 \times 8 = 40$

1. (a) What is the toggling problem in latches? How does the T latch resolve this issue? CLO3 2
(b) Compare and contrast latches with flip-flops in terms of memory elements, timing dependency, and clock signal requirements. CLO3 3
(c) Explain the working principle of an JK Latch. Construct its truth table and state the behavior for each combination of inputs. CLO3 3
2. (a) Explain the difference between edge-triggered and level-triggered flip-flops. CLO3 2
(b) Explain the race condition that occurs in a JK Flip-Flop and how it can be avoided. CLO3 3
(c) Design a master-slave JK Flip-Flop using two JK Flip-Flops and explain its operation. CLO1 3
3. (a) Compare the conversions between SR, JK, D, and T Flip-Flops. Explain the situations where these conversions are useful in digital systems. CLO4 4
(b) Demonstrate how an SR Flip-Flop can be converted to a JK Flip-Flop using NAND gates. CLO4 4
4. (a) Perform state reduction on the following state diagram. Draw the reduced state diagram. CLO1 4



- (b) Design a 4 bit asynchronous UP/DOWN counter with logical explanation of your design. CLO5 4

6. (a) Briefly explain the fundamental issues relating to the design of instruction sets. CLO5 4
- (b) Describe different status flags used in the x86 processors. CLO5 4
7. (a) Illustrate a generic model of an I/O module. State different types of I/O commands that an I/O module may receive. CLO1 4
- (b) Draw the flowchart for input of a block of data from a peripheral device in an interrupt-driven I/O module. CLO1 4



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 229 : Computer Architecture

Batch: 24th

Date: 22/03/2025

Final Examination

Semester: Fall 2024

Time: 2 Hours

Total Marks: 40

Answer any five questions.

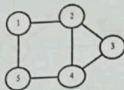
5 × 8 = 40

- | | | | | |
|----|-----|--|------|-------|
| 1. | (a) | Draw the flowchart of Booth's algorithm for twos complement integer multiplication. | CLO4 | 4 |
| | (b) | Perform the Twos Complement Multiplication between the 4-bit signed binary numbers (0111) ₂ (Multiplicand) and (0011) ₂ (Multiplier) using Booth's algorithm. Show the change in values of the accumulator (A), multiplier (Q) and multiplicand (M) register after each cycle. | CLO4 | 4 |
| 2. | (a) | Mention different characteristics of a computer memory system. | CLO1 | 2 |
| | (b) | Briefly describe the key performance parameters for memory. | CLO1 | 3 |
| | (c) | Consider a two-level cache memory where L1 cache has an access time of 0.02 ms with a hit ratio of 90% and L2 cache has an access time of 0.2 ms. Now, calculate the average memory access time for this system. | CLO1 | 3 |
| 3. | (a) | Briefly describe direct and associative mapping functions and illustrate how the mapping works. | CLO3 | 4 |
| | (b) | Suppose, a direct-mapped cache has 5 cache lines. If memory blocks are numbered 0, 1, 2, 3, 4, 5, ... then determine the mapping of memory blocks to cache lines. | CLO3 | 4 |
| 4. | (a) | What are cache replacement algorithms? Briefly describe the different types of replacement algorithms. | CLO3 | 4 |
| | (b) | Illustrate a computer system that uses (i) logical cache or (ii) physical cache. | CLO3 | 4 |
| 5. | (a) | Define the following –
i. Machine Instruction
ii. Instruction Set
iii. Opcode | CLO2 | 3×1=3 |
| | (b) | Provide a simple example of an instruction format and draw the state diagram for the instruction cycle showing the steps involved. | CLO2 | 5 |

8 9 12 13 5 2 1
3 9 4 4 10 5



4. (a) Define overlapping sub-problem. How memorization improve the performance in solving overlapping sub-problem? CLO5 3
- (b) Consider the following instance of the 0-1 knapsack problem:
 Number of items, $n = 4$,
 Capacity, $W = 13$,
 Profits $\{p_1, p_2, p_3, p_4\} = \{7, 18, 8, 9\}$ and
 Weights $\{w_1, w_2, w_3, w_4\} = \{4, 7, 5, 2\}$
 Calculate the optimal solution using dynamic programming. CLO5 5
5. (a) Distinguish between backtracking and recursion. CLO2 2
- (b) Find out three possible solutions for coloring the following graph in RGB color mode: CLO2 6



6. (a) Apply branch-and-bound technique to solve 15-puzzle problem for the particular organizations: CLO5 4

1	2	3	4
5	6	7	
9	10	12	8
13	14	11	15

- (b) Let you have four jobs with the following penalties, deadlines and required times respectively: CLO5 4
- $\{p_1, p_2, p_3, p_4\} = \{5, 10, 6, 3\}$,
 $\{d_1, d_2, d_3, d_4\} = \{1, 3, 2, 1\}$ and
 $\{t_1, t_2, t_3, t_4\} = \{1, 2, 1, 1\}$
 Now, generate the state space tree to sequence the jobs with deadlines using branch and bound approach.

7. (a) Define Lower Bound Theory. Mention its types. CLO3 2
- (b) Write the advantages and disadvantages of Lower Bound Theory. CLO3 3
- (c) Prove that any decision tree that can sort n elements must have the height $\Omega(n \log n)$. CLO1 3

Solu $= \{G, R, G, B, R\}$
 $= \{R, G, R, B, G\}$
 $= \{R, B, R, B, G\}$



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE221: Computer Algorithms

Final Examination

Semester: Fall 2024

Batch: 24th

Date: 13-03-2025

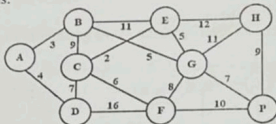
Time: 2 Hours
Total Marks: 40

CLOs Marks

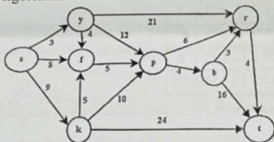
CLO5 4

CLO5 4

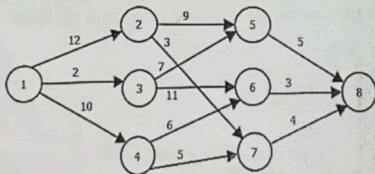
1. (a) State Kruskal's Algorithm.
(b) Apply Prim's Algorithm to generate the minimum spanning tree for the following graphs:



2. (a) Can greedy algorithms be used to solve optimization problems? Give an example.
(b) Assume vertex 's' is the source of the following graph, calculate the shortest path using Dijkstra's Algorithm:



3. (a) You are given a directed weighted graph and the shortest path from a source vertex 's' to a destination vertex 'p'. If weight of every edge is increased by 5 units, does the shortest path remain same in the modified graph? Depend your opinion with an example.
(b) Find the minimum cost from node 1 to node 8 for the following multistage graph using dynamic programming technique:



for the given segment.

7. a) Explain two usages of the keyword **static** in OOP.

CLO1

3

b) There is a String "**Hello Gen Z**" and you need to reverse it. You have to show two ways of how to reverse the String by using build-in methods of Java and doing manually.

CLO3

5



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE 223: Object Oriented Programming

Batch: 24th

Date: 15/03/2025

Final Examination

Semester: Fall 2024

Time: 2 Hours

Total Marks: 40

5 × 8 = 40

Answer **any five** questions of the followings.

		CLOs	Marks
1. a)	What is Java Exception Handling? Briefly explain the key terms of Java Exception Handling.	CLO1	4
b)	Explain <u>ArithmeticException</u> and <u>ArrayIndexOutOfBoundsException</u> with example.	CLO5	4
2. a)	"In Java, strings are immutable." Justify the given sentence.	CLO1	4
b)	Write a program in Java to find a string is Palindrome or not.	CLO5	4
3. a)	Briefly explain Polymorphism using method overriding.	CLO2	3
b)	Write a Java program to create and display a unique three-digit number using 1, 2, 3 and 4. Also count how many three-digit numbers are there.	CLO3	5
4. a)	Define Abstract Window Toolkit (AWT) in Java.	CLO4	2
b)	Write a Java program to create a class called Point with instance variables x and y. Implement overloaded constructors: - One constructor takes int parameters - Another constructor takes double parameters - Print the values of the variables for each constructor	CLO2	6
5. a)	What is multithreading in Object Oriented Programming?	CLO1	2
b)	Write a Java program to create a class Person with methods called getFirstName() and getLastName() . Create a subclass called Employee that adds a new method named getEmployeeId() and overrides the getLastName() method to include the employee's job title.	CLO3	6
6. a)	Define Variable-length arguments in Object Oriented Programming.	CLO1	2
b)	Consider the following code segment:	CLO2	6

```
public class VarExp {
    public static void main(String[] args) {
        int a = minmax("min", 2, 1, 6, 4, 5); // a = 1
        int b = minmax("min", 3, 0, 6); // b = 0
        int c = minmax("max", 1, 2, 6, 5); // c = 6
        int d = minmax("max", 1, 3, 7); // d = 7
    }
}
```

Write the Java method minmax. You can write only one minmax method

void
public static



STANDARD UNIVERSITY BANGLADESH

Faculty Science, Engineering and Technology
Department of Computer Science and Engineering

ENS221: Environmental Science

Final Examination

Semester: Fall 2024
Time: 2 Hours
Total Marks: 40

Date: 17/03/25

<u>Part A (Answer any One)</u>		CLO	Marks
1. a) Definition of climate smart agriculture and explain three indicators of it.		CLO2	3.5
b) Provide a number of expert opinions for preventing the damage of flash flood in Bangladesh		CLO2	6.5
2. a) What is the main idea of drought? Explain the features of it in the context of Bangladesh.		CLO2	5
b) State the threats and safety measures of lightning and thunderstorm in the context of Bangladesh.		CLO2	5
<u>Part B (Answer any Three)</u>			
3. a) Explain mitigation and low carbon development.		CLO3	3
b) Write Six Pillars of Bangladesh Climate Change Strategy Action Plan and explain the food security social protection and health.		CLO3	7
4. a) Analyze how the primary indications of eco-tourism can help to ensure sustainability of it.		CLO3	5
b) Calculate how much a mango tree sequesters carbon in a year. A 15-year-old mango tree would probably grow about <u>22</u> feet tall with a trunk about <u>13</u> inches in diameter.		CLO3	5
5. a) Analyze that a number of environmental impacts of landslide in the light of Bangladesh.		CLO3	5
b) Evaluate the economic loss and damage of landslide.		CLO3	5
6. a) What do you understand by sedimentation process?		CLO3	2
b) Evaluate the different types of process of sedimentation in the context of Bangladesh (with figure).		CLO3	8